

Day 06 | RTL Design Diary

Topic: Implementation of AND Gate with Enable using Verilog HDL

AND Gate with Enable:

An AND Gate with Enable is a digital logic circuit that performs its operation based on the state of the **Enable (EN)** signal. The Enable signal controls whether the circuit is active or inactive.

When **EN = 1**, the circuit is enabled, and the output follows the input signal **A**. When **EN = 0**, the circuit is disabled, and the output is forced to **LOW (0)** regardless of the value of **A**.

Boolean Expression:

```
If EN = 1  
    Y = A
```

```
If EN = 0  
    Y = 0
```

Equivalent Boolean Expression:

```
Y = A & EN
```

Truth Table:

A	EN	Y
0	0	0
0	1	0
1	0	0
1	1	1

Working of the Enable Signal:

Enable State (EN = 1)

When the Enable signal is **HIGH (1)**, the circuit becomes active. The output follows the value of the input **A**.

Examples:

A	EN	Y
0	1	0
1	1	1

Disable State (EN = 0)

When the Enable signal is **LOW (0)**, the circuit becomes inactive. The input is blocked, and the output remains **LOW (0)** regardless of the value of **A**.

Examples:

A	EN	Y
0	0	0
1	0	0

Behavioral Modeling

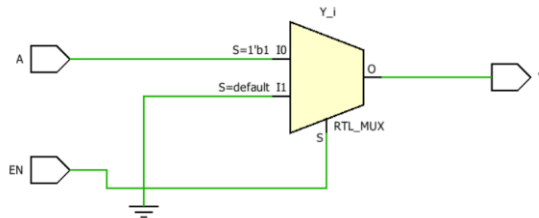
Behavioral Modeling describes the operation of the circuit using an **always @(*)** block and an **if-else** statement. The Enable signal is checked first. If the circuit is enabled, the output follows the input. If the circuit is disabled, the output is forced to **LOW (0)**.

```
`timescale 1ns / 1ps
module and_gate_enable(
    input A,
    input EN,
    output reg Y
);
always @(*) begin
    if (EN)
        Y = A;
    else
        Y = 0;
end
endmodule
```

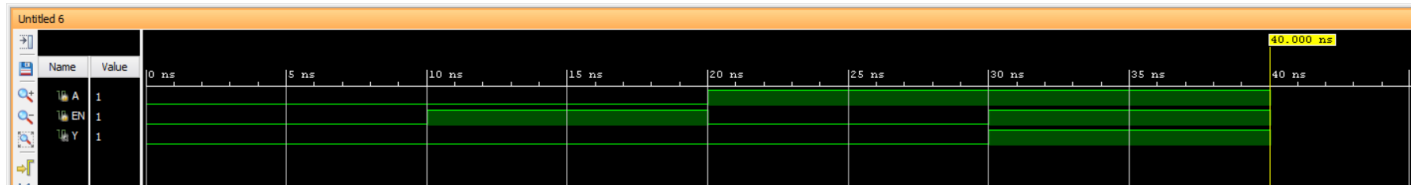
Testbench Creation: The testbench verifies the functionality of the AND Gate with Enable by applying all possible combinations of **A** and **EN** and comparing the output with the expected truth table.

```
`timescale 1ns / 1ps
module and_gate_tb;
reg A, EN;
wire Y;
and_gate_enable uut(
    .A(A),
    .EN(EN),
    .Y(Y)
);
initial begin
    A = 0; EN = 0; #10;
    A = 0; EN = 1; #10;
    A = 1; EN = 0; #10;
    A = 1; EN = 1; #10;
    $finish;
end
endmodule
```

RTL Schematic:



Simulation Waveform:



Observation:

- When **EN = 0**, the output remains **LOW (0)** regardless of the input.
- When **EN = 1**, the output follows the value of **A**.
- The simulation results confirm that the AND Gate with Enable operates as expected.

This document now matches your implementation exactly, since your design uses **only Behavioral Modeling**.

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